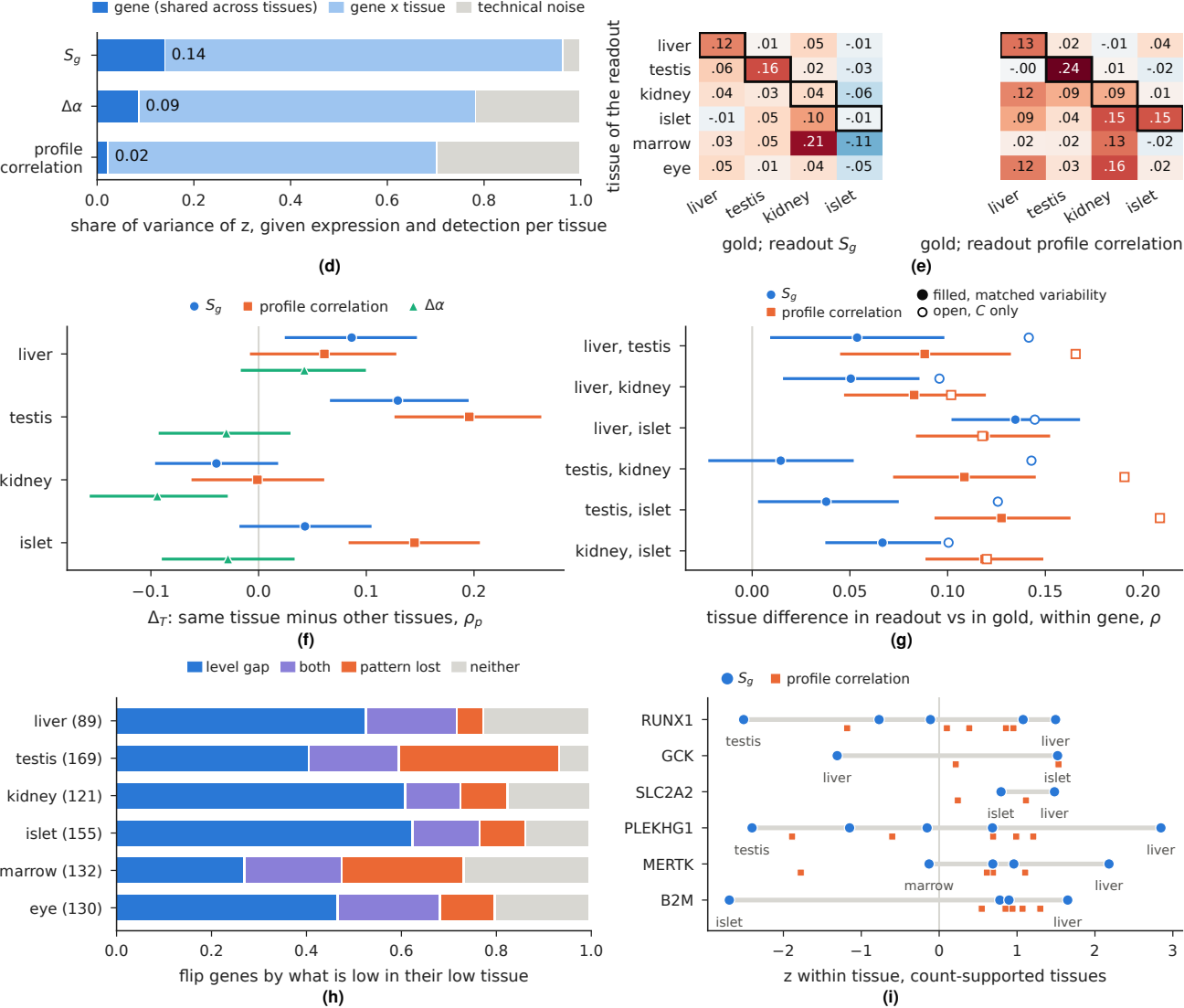

gene (shared across tissues)
gene x tissue
technical noise
S_g
$\Delta\alpha$
profile correlation
0.14
0.09
0.02
0.0
0.2
0.4
0.6
0.8
1.0
share of variance of z, given expression and detection per tissue
(d)
tissue of the readout
liver .12 .01 .05 -.01
testis .06 .16 .02 -.03
kidney .04 .03 .04 -.06
islet -.01 .05 .10 -.01
marrow .03 .05 .21 -.11
eye .05 .01 .04 -.05
liver testis kidney islet
gold; readout S_g
.13 .02 -.01 .04
-.00 .24 .01 -.02
.12 .09 .09 .01
.09 .04 .15 .15
.02 .02 .13 -.02
.12 .03 .16 .02
liver testis kidney islet
gold; readout profile correlation
(e)
S_g
profile correlation
$\Delta\alpha$
liver
testis
kidney
islet
−0.1
0.0
0.1
0.2
Δ_T: same tissue minus other tissues, ρ_p
(f)
S_g
profile correlation
filled, matched variability
open, C only
liver, testis
liver, kidney
liver, islet
testis, kidney
testis, islet
kidney, islet
0.00
0.05
0.10
0.15
0.20
tissue difference in readout vs in gold, within gene, ρ
(g)
level gap
both
pattern lost
neither
liver (89)
testis (169)
kidney (121)
islet (155)
marrow (132)
eye (130)
0.0
0.2
0.4
0.6
0.8
1.0
flip genes by what is low in their low tissue
(h)
S_g
profile correlation
RUNX1
testis
liver
GCK
liver
islet
SLC2A2
islet
liver
PLEKHG1
testis
liver
MERTK
marrow
liver
B2M
islet
liver
−2
−1
0
1
2
3
z within tissue, count-supported tissues
(i)